Lec23

26 February 2026 09:20 AM

[1, 2, 3, 1] k = 3

0 1 2 3

i j

[0, 3]

-3 = i-j

3 <= 3

4 <= 3

[1, 2, 3, 1]

i j j j

0 3

nums[i] == nums[j]

-3 = i-j

3 <= k

return [i, j]

[1, 2, 3, 1, 2, 3] k = 2

j

j

nums[i] = 1 == 3 !

-3 = i-j

3 <= 2

2 == 3

2 == 2

3 == 3

3 = 1-4

3 = 2-5

i j

3 = 5-2

Math.abs(i-j)

j-i

```
for(i=0 ; i<n-1 ; i++){
    for (j=i+1 ; j<n ; j++){
        if (nums[i] == nums[j] && i-j <= k){
```

```
for (j=i+1; j<=i+k; j++){
    if (nums[i] == nums[j] && i-j <= k){
        return true
    }
}
}
return false
```

31, next permutation

[1, 2, 3]

n!

1 2 3
1 3 2
2 1 3
2 3 1
3 1 2
3 2 1

3

1 2 3
1 3 2

[1, 2, 4, 3, 7, 6, 5]

n-2

num[i] < num[i+1]

6 < 7
5 < 6

1, 2, 4, 3, 6 7

pivot = 2

① find pivot element

② find greater

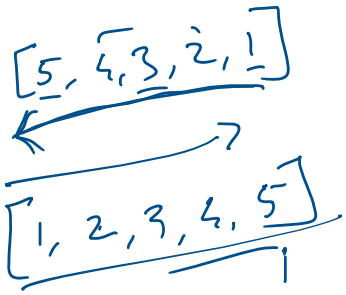
③ swap greater pivot

④ reverse pivot+1-n

[5, 4, 3, 2, 1]

[1, 2, 3, 6, 5, 4]

[1, 2, 3, 6, 4, 5]



④ reverse pivot+1-n

pivot = ~~-1~~ 3

6 < 4
5 < 6

5 < 4
5 < 6

pivot = -1

```
① for(i = n-2; i>=0; i--){
      if(nums[i] < nums[i+1]){
          pivot = i
          break;
      }
if (pivot == -1){
    reverse( )
    return
}

for(i = n-1; i > pivot; i--){
    if(nums[i] > nums[pivot])
        swap ( )
        break

}

reverse(pivot+1, n-1)
```